19-202-0706 BIOMETRIC TECHNOLOGIES

Course Outcomes:

On completion of this course the student will be able to:

1. *Explain the need and different types of biometric systems*
2. *Describe the characteristics and deployment of systems using physical biometric systems*
3. *Describe the characteristics and deployment of systems using behavioural biometric systems*
4. *Describe the characteristics and deployment of different biometric interfaces.*
5. *Analyse different applications and suggest the suitable biometric system for a given application.*

Module I

Biometrics: Introduction - Biometrics versus traditional techniques - Characteristics - Key biometric processes - Performance measures in biometric systems - Assessing the privacy risks of biometrics - Different biometric standards - Designing and Selecting a biometric system - Application - Key biometric terms and processes - Biometric matching methods - Accuracy in biometric system.

Module II

Physical Biometrics: Fingerprints - Technical description - Characteristics - Competing technology - Strengths, Weaknesses, Deployments. Facial scan technical description - Characteristics, Weaknesses, Deployments - Retina vascular pattern Technical description - Characteristics, Strength, Weaknesses, Deployments - DNA, Facial scan, Ear scan, Retina scan, Iris scan, Automated fingerprint identification system, Palm print, Hand vascular geometry analysis, Knuckle, Dental, Cognitive Biometrics - ECG, EEG.

Module III

Behavioural Biometrics: Handprint biometrics - Signature and handwriting technology - Technical description – Classification - Comprehensive packet logging - Keyboard or keystroke dynamics - Voice, Data acquisition, Feature extraction - Characteristics, Strength, Weakness, Deployment. Gait recognition, Gesture recognition, Video face, Mapping the body technology. Multi Biometrics and Multi Factor biometrics - Two factor authentication with password - Tickets and tokens

Module IV

USER INTERFACES: Biometric interfaces: Human machine interface - BHMI structure, Human side interface: Iris image interface - Hand geometry and fingerprint sensor - Executive decision - Implementation plan - Case study on physiological, Behavioural and multi factor biometrics in identification system. Categorizing biometric applications, Application areas: Criminal and citizen identification – Surveillance - PC/network access - E-commerce and retail/ATM, Costs to deploy, Issues in deployment, Biometrics in medicine, cancellable biometrics.

References:

1. Samir Nanavati, Michael Thieme and Raj Nanavati, “Biometrics – Identity Verification in a Networked World”, John Wiley and Sons, New Delhi, 2003.
2. Anil K Jain, Patrick Flynn and Arun A Ross, “Handbook of Biometrics”, Springer, USA, 2010.
3. John R Vacca, “Biometric Technologies and Verification Systems”, Elsevier, USA, 2009.
4. David Chek Ling Ngo, Andrew Beng Jin Teoh, Jiankun Hu, “Biometric Security”, Cambridge Scholars Publishing, 2015.
5. Paul Reid, “Biometrics for Network Security, Pearson Education, New Delhi, 2004.
6. Julian Ashbourn,” Guild to Biometric for Large Scale System: Technological, Operational and User Related Factor”, Springer Data London Limited, 2011